

Reverse an Array - TCS Style Explanation

This document explains how to reverse an array using Python in a simple TCS interview style.

Problem Statement: Reverse the given array.

Input:
[1, 2, 3, 4, 5]

Output:
[5, 4, 3, 2, 1]

Logic:

We use two pointers: one from the beginning and one from the end. Swap the elements and move both pointers toward the center until they meet.

Python Program:

```
# Reverse an Array

arr = [1, 2, 3, 4, 5]

left = 0
right = len(arr) - 1

# Swap elements
while left < right:

    arr[left], arr[right] = arr[right], arr[left]

    left += 1
    right -= 1

# Print reversed array
print(arr)
```

Dry Run:

Original Array:
[1, 2, 3, 4, 5]

Step 1:
Swap 1 and 5
[5, 2, 3, 4, 1]

Step 2:
Swap 2 and 4
[5, 4, 3, 2, 1]

Final Output:
[5, 4, 3, 2, 1]